RULLMR Session 03: Data collection with a local LLM

Preparation for 21 October 2026

Next week we focus on common tasks in **data collection** with a bit of best **data management** practices**.** This session will be much more focused on practicing using LLMs in-class compared to the previous one. **You do not need to finish the homework or arrive with a working workflow.** You have plenty of time during the session to develop this in pairs, and we will demonstrate some possible solutions in-class. Still, it is useful to experiment a bit with the homework yourself so that at a minimum, you're familiar with the tasks and can explicate possible solutions. All the homework materials can be found under week 3 at https://bma-vandijk.github.io/RULLMR/schedule/.

Task 1
A common problem is encountering messy, inconsistent data in tables. We have a mock table called noisy_ehr.csv that you should clean and standardize. That means, finding a common format for dates, age, id, etc. Before you begin, try to think of a simple workflow where you reflect on your goal, context, and evaluation. Also think about logging info that makes your work as reproducible as possible. You can use LLMs to design this workflow. By now you know you can use LLMs in different environments, and you also know that each environment comes with different pros and cons.

Use a LLM to design different approaches to do the task, execute, and create a brief report listing your findings, as well as the pros and cons for every approach (e.g. performance, speed, privacy, control, reproducibility, etc.). Include info on the model, version, prompt, model settings, etc.

For the curious budding programmer: try to generate and execute a Python script to do the task. What are the pros and cons of this and when do you think this is most useful?

Task 2 (additional challenge)
The reports_homework_10_std_chatgpt.csv file contains 10 doctors' notes from real LUMC emergency department visits. Potentially identifying and clinical details have been substantially altered. The notes retain abbreviations, fragments, spelling mistakes, missing information and occasional inconsistencies. We keep the notes in Dutch to reflect everyday clinical documentation at LUMC. This adds a realistic challenge when extracting information with an LLM.

Try to design a workflow that uses a local LLM to extract data from all the notes. As for task 1, choose your own model and tools. Aim for one workflow that can process every report without manually rewriting the prompt for each case.

Some tips:

1. Choose a small set of variables, presenting complaint, laboratory results, diagnosis and disposition. Think about how to handle missing, unclear or contradictory information.
2. Try your approach on a few of the 10 supplied reports. When you have output, check the extracted values against the notes and improve your approach.
3. If you get this far, save your results as a structured table, keeping the case_id so each result links to its teaching note.

As for Task 1, document what you try: the model, version, prompt(s), relevant settings such as temperature and context length, and the steps of your workflow. Note what worked, what did not, and any changes you made. Bring whatever you have: ideas, prompts, partial results or questions. If you get stuck, note where it happened.

In class, you can test your workflow on the complete dataset of 36 reports, including 26 reports you have not received beforehand.

Good luck!